

```
# make install-commandmode
/usr/bin/install -c -m 775 -o nagios -g nagcmd -d /usr/local/
nagios/var/rw
chmod g+s /usr/local/nagios/var/rw
*** External command directory configured ***
```

Configure Nagios for Apache:

```
# sudo make install-webconf
/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/apache2/
conf.d/nagios.conf
*** Nagios/Apache conf file installed ***
```

Set a Web admin password for Nagios:

```
# htpasswd2 -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password: (admin)
Re-type new password:
Adding password for user nagiosadmin
```

Next, restart Apache and check its status with `/etc/init.d/apache2 restart` and `service apache2 status` respectively.

Now download the Nagios plug-in from <http://www.nagios.org/download/plugins/> and extract the file; then issue the following command:

```
# cd nagios-plugins-1.4.15
# ./configure --with-user=nagios --with-group=nagcmd
# make
```

Wait for approximately 10-15 minutes till the command runs, then finish it with a `sudo make install` command. Add Nagios to the list of system services to ensure it automatically activates on system start:

```
chkconfig --add nagios
chkconfig nagios on
```

Verify the sample Nagios configuration files as follows:

```
/usr/local/nagios/bin/nagios -v /usr/local/nagios/etc/nagios.cfg
```

If there are no errors, start Nagios with `service nagios start`, then open a browser and access <http://localhost/nagios> and log in as `nagiosadmin` with the password you set.

Nagios Checker (Firefox add-on)

The Nagios Checker Mozilla Firefox add-on provides the status-bar display of information on the status of resources from Nagios. The Nagios Web interface information is parsed to make information available to the indicator. The add-on has been verified with Nagios 3, 2.5+, 2.0b4, 1.5, 1.3 and 1.2. You can add it from <https://addons.mozilla.org/en-US/firefox/addon/nagios-checker/>. After restarting Firefox,

right-click the 'N' sign in the status-bar and click *Settings*; provide the Nagios Web URL, the username and password (Figure 2). Provide the *Status Script URL*, as seen in Figure 3. Click OK, and within seconds, you will see the Nagios status on the status bar. Click on it and a detailed list will pop up (Figure 4).

Cloud monitoring with Nagios

As per NIST's definition, cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources such as networks, servers and storage, which can be rapidly provisioned and de-provisioned with minimal management effort.

Virtualisation is the core of cloud computing, since it provides cost savings by reducing up-front investment in infrastructure.

Cloud monitoring refers to monitoring resources provided by cloud service providers. In the virtual world, this is not just necessary but crucial for high availability and fault tolerance, and to avoid single points of failure, since resources are shared. Monitoring of virtual systems is dynamic, since the resources are virtual and can be treated like files. Resource monitoring for scaling up and scaling down is also critical considering the huge capacity offered by cloud service providers.

Public cloud monitoring

Cloud monitoring refers to the monitoring of the performance of physical or virtual servers, storage, networks and the applications running on them. Cloud monitoring tools are used to collect data and illustrate patterns that might otherwise be difficult to spot in dynamic infrastructure. Nagios provides monitoring for cloud resources—compute, storage and network services. Nagios is proficient at monitoring a variety of operating systems to detect cloud environment issues, network outages and application availability.

Amazon Web Services (AWS) monitoring

Nagios solutions that provide cloud monitoring capabilities (for Amazon EC2, Amazon Simple Storage Services, etc) are Nagios XI and Nagios Core. 

References

- [1] Nagios Core documentation:
<http://nagios.sourceforge.net/docs/nagioscore/3/en/toc.html>
- [2] http://complanetworking.about.com/od/itinformationtechnology/t/net_monitoring.htm
- [3] <http://library.nagios.com/library/products/nagiosxi/manuals>
- [4] <https://addons.mozilla.org/en-US/firefox/addon/nagios-checker/>
- [5] <http://searchenterpriselinix.techtarget.com/definition/Nagios>

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